Inverse Trigonometry Function

71. Considering the principal values of the inverse trigonometric functions, $\sin^{-1}\left(\frac{\sqrt{3}}{2}x+\frac{1}{2}\sqrt{1-x^2}\right), -\frac{1}{2}<x<\frac{1}{\sqrt{2}}$, is equal to :

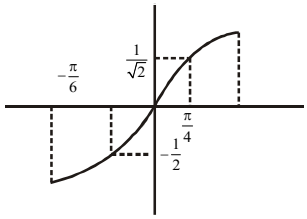
प्रतिलोम त्रिकोणमितीय फलनों के मुख्य मान लेने पर $\sin^{-1}\left(\frac{\sqrt{3}}{2}x+\frac{1}{2}\sqrt{1-x^2}\right), -\frac{1}{2}<x<\frac{1}{\sqrt{2}}$ का मान होगा।

(A) $\frac{5\pi}{6}-\sin^{-1}x$ (B) $\frac{\pi}{4}+\sin^{-1}x$ (C) $\frac{-5\pi}{6}-\sin^{-1}x$ (D) $\frac{\pi}{6}+\sin^{-1}x$

Ans. D

Sol. $\sin^{-1}\left(\frac{\sqrt{3}}{2}x+\frac{1}{2}\sqrt{1-x^2}\right); -\frac{1}{2}<x<\frac{1}{\sqrt{2}}$

Let $x=\sin\theta$



$-\frac{\pi}{6}<\theta<\frac{\pi}{4}$

$\sin^{-1}\left[\frac{\sqrt{3}}{2}\sin\theta+\frac{1}{2}\cos\theta\right]$

$\sin^{-1}\left[\cos\frac{\pi}{6}\sin\theta+\sin\frac{\pi}{6}\cos\theta\right]$

$\sin^{-1}\left(\sin\left(\theta+\frac{\pi}{6}\right)\right)$

$0<\theta+\frac{\pi}{6}<\frac{\pi}{4}+\frac{\pi}{6}$

$0<\theta+\frac{\pi}{6}<\frac{5\pi}{12}$

$\Rightarrow \theta+\frac{\pi}{6}$

$\Rightarrow \frac{\pi}{6} + \sin^{-1} x$

72. Let the domains of the function $f(x) = \log_4 \log_3 \log_7 \left(8 - \log_2 \left(x^2 + 4x + 5\right)\right)$ and $g(x) = \sin^{-1}\left(\frac{7x+10}{x-2}\right)$ be (α, β) and $[\gamma, \delta]$, respectively. Then $\alpha^2 + \beta^2 + \gamma^2 + \delta^2$ is equal to :

माना फलनों $f(x) = \log_4 \log_3 \log_7 \left(8 - \log_2 \left(x^2 + 4x + 5\right)\right)$ और $g(x) = \sin^{-1}\left(\frac{7x+10}{x-2}\right)$ के प्रान्त क्रमशः (α, β) और $[\gamma, \delta]$ हैं। तो $\alpha^2 + \beta^2 + \gamma^2 + \delta^2$ बराबर है :

(A) 13 (B) 15 (C) 14 (D) 16

Ans. B

Sol. $f(x) = \log_4 \log_3 \log_7 (8 - \log_2(x^2 + 4x + 5))$

$\Rightarrow x^2 + 4x + 5 > 0 \ \forall \ x \in R$

$\Rightarrow 8 - \log_2(x^2 + 4x + 5) > 0$ ...(1)

$\Rightarrow \log_7(8 - \log_2(x^2 + 4x + 5)) > 0$

$= 8 - \log_2(x^2 + 4x + 5) > 1$ ...(2)

$\Rightarrow \log_3 \log_7 (8 - \log_2(x^2 + 4x + 5)) > 0$

$= 8 - \log_2(x^2 + 4x + 5) > 7$ ...(3)

from (1), (2), (3)

$\log_2(x^2 + 4x + 5) < 1$

$x^2 + 4x + 5 < 2$

$x^2 + 4x + 3 < 0$

$(x + 1)(x + 3) < 0 \Rightarrow x \in (-3, -1)$

$g(x) = \sin^{-1}\left(\frac{7x+10}{x-2}\right)$

$-1 \le \frac{7x+10}{x-2} \le 1$

$\frac{7x+10}{x-2} + 1 \ge 0$ & $\frac{7x+10}{x-2} - 1 \le 0$

$\frac{8x+8}{x-2} \ge 0$ $\frac{6x+12}{x-2} \le 0$

$x \in (-\infty, -1] \cup (2, \infty)$ $x \in [-2, 2)$

$D_g : [-2, -1]$

$\alpha^2 + \beta^2 + \gamma^2 + \delta^2 = 9 + 1 + 4 + 1 = 15$

73. The sum of the infinite series $\cot^{-1}\left(\frac{7}{4}\right)+\cot^{-1}\left(\frac{19}{4}\right)+\cot^{-1}\left(\frac{39}{4}\right)+\cot^{-1}\left(\frac{67}{4}\right)+\ldots$ is :

अनंत श्रेणी $\cot^{-1}\left(\frac{7}{4}\right)+\cot^{-1}\left(\frac{19}{4}\right)+\cot^{-1}\left(\frac{39}{4}\right)+\cot^{-1}\left(\frac{67}{4}\right)+\ldots$ का योग है :

(A) $\frac{\pi}{2}-\cot^{-1}\left(\frac{1}{2}\right)$ (B) $\frac{\pi}{2}-\tan^{-1}\left(\frac{1}{2}\right)$ (C) $\frac{\pi}{2}+\tan^{-1}\left(\frac{1}{2}\right)$ (D) $\frac{\pi}{2}+\cot^{-1}\left(\frac{1}{2}\right)$

Ans. B

Sol. $\tan^{-1}\left(\frac{4}{7}\right)+\tan^{-1}\left(\frac{4}{19}\right)+\tan^{-1}\left(\frac{4}{39}\right)+....$

$S = 7 + 19 + 39 + 67 + ... + T_n$

$S = \quad 7 + 19 + 39 + ... + T_n$

–

$T_n = 7 + (12 + 20 + 28 + ... (n - 1) \text{ terms})$

$T_n = 7+\frac{n-1}{2}\left(24+(n-2)8\right)$

$T_n = 7 + (n - 1)(4 + 4n)$

$T_n = 7 + 4(n^2 - 1) = 4n^2 + 3$

$$\sum_{n=1}^{\infty}\tan^{-1}\left(\frac{4}{4n^2+3}\right)=\sum_{n=1}^{\infty}\tan^{-1}\left(\frac{1}{n^2+\frac{3}{4}}\right)$$

$$\Rightarrow\sum_{n=1}^{\infty}\tan^{-1}\left(\frac{\left(n+\frac{1}{2}\right)-\left(n-\frac{1}{2}\right)}{1+\left(n-\frac{1}{2}\right)\left(n+\frac{1}{2}\right)}\right)$$

$$\Rightarrow\sum_{n=1}^{\infty}\left(\tan^{-1}\left(n+\frac{1}{2}\right)-\tan^{-1}\left(n-\frac{1}{2}\right)\right)$$

$$\Rightarrow\frac{\pi}{2}-\tan^{-1}\frac{1}{2}$$

74. Using the principal values of the inverse trigonometric functions, the sum of the maximum and the minimum values of $16((\sec^{-1}x)^2 + (\text{cosec}^{-1}x)^2)$ is :

प्रतिलोम त्रिकोणमितीय फलनों के मुख्य मानों का प्रयोग करते हुए $16((\sec^{-1}x)^2 + (\text{cosec}^{-1}x)^2)$ के अधिकतम और न्यूनतम मानों का

योग है :

(A) $31\pi^2$ (B) $24\pi^2$ (C) $22\pi^2$ (D) $18\pi^2$

Ans. C

Sol. $16(\sec^{-1}x)^2 + (\text{cosec}^{-1}x)^2$

$\sec^{-1} x = a \in [0,\pi] - \left\{\frac{\pi}{2}\right\}$

$\text{cosec}^{-1}x = \frac{\pi}{2} - a$

$= 16\left[a^2 + \left(\frac{\pi}{2} - a\right)^2\right] = 16\left[2a^2 - \pi a + \frac{\pi^2}{4}\right]$

$\max]_{a=\pi} = 16\left[2\pi^2 - \pi^2 + \pi\frac{2}{4}\right] = 20\pi^2$

$\min]_{a=\frac{\pi}{4}} = 16\left[\frac{2\times\pi^2}{16} - \frac{\pi^2}{4} + \frac{\pi^2}{4}\right] = 2\pi^2$

Sum = $22\pi^2$

75. If $\frac{\pi}{2} \le x \le \frac{3\pi}{4}$, then $\cos^{-1}\left(\frac{12}{13}\cos x + \frac{5}{13}\sin x\right)$ is equal to :

यदि $\frac{\pi}{2} \le x \le \frac{3\pi}{4}$ है, तो $\cos^{-1}\left(\frac{12}{13}\cos x + \frac{5}{13}\sin x\right)$ बराबर है :

(A) $x - \tan^{-1}\frac{5}{12}$ (B) $x + \tan^{-1}\frac{4}{5}$ (C) $x + \tan^{-1}\frac{5}{12}$ (D) $x - \tan^{-1}\frac{4}{3}$

Ans. A

Sol. $\cos^{-1}(\cos\alpha\cos a + \sin\alpha\sin x)$ $\cos\alpha = \frac{12}{13}$

$\cos^{-1}(\cos(x-\alpha))$ $\sin\alpha = \frac{5}{13}$

$= x - \alpha$ $\tan\alpha = \frac{5}{12}$

$= x - \tan^{-1}\left(\frac{5}{12}\right)$

76. If $\alpha > \beta > \gamma > 0$, then the expression $\cot^{-1}\left\{\beta+\frac{(1+\beta^2)}{(\alpha-\beta)}\right\}+\cot^{-1}\left\{\gamma+\frac{(1+\gamma^2)}{(\beta-\gamma)}\right\}+\cot^{-1}\left\{\alpha+\frac{(1+\alpha^2)}{(\gamma-\alpha)}\right\}$ is equal to :

यदि $\alpha > \beta > \gamma > 0$ है, तो व्यंजक $\cot^{-1}\left\{\beta+\frac{(1+\beta^2)}{(\alpha-\beta)}\right\}+\cot^{-1}\left\{\gamma+\frac{(1+\gamma^2)}{(\beta-\gamma)}\right\}+\cot^{-1}\left\{\alpha+\frac{(1+\alpha^2)}{(\gamma-\alpha)}\right\}$ बराबर है :

(A) 0 (B) π (C) $\frac{\pi}{2}-(\alpha+\beta+\gamma)$ (D) 3π

Ans. B

Sol. $\cot^{-1}\left(\frac{\alpha\beta+1}{\alpha+\beta}\right)+\cot^{-1}\left(\frac{\beta\gamma+1}{\beta-\gamma}\right)+\cot^{-1}\left(\frac{\gamma\alpha+1}{\gamma-\alpha}\right)$

$\Rightarrow \tan^{-1}\left(\frac{\alpha-\beta}{1+\alpha\beta}\right)+\tan^{-1}\left(\frac{\beta-\gamma}{1+\beta\gamma}\right)+\left(\pi+\tan^{-1}\left(\frac{\gamma-\alpha}{1+\gamma\alpha}\right)\right)$

$\Rightarrow \tan^{-1}(\alpha)-\tan^{-1}\beta+\tan^{-1}\beta-\tan^{-1}\gamma+\pi+\tan^{-1}\gamma-\tan^{-1}\alpha=\pi$

77. Given that the inverse trigonometric function assumes principal values only. Let x, y be any two real numbers in [–1,1] such that $\cos^{-1}x-\sin^{-1}y=\alpha$, $\frac{-\pi}{2}\le\alpha\le\pi$. Then the minimum value of $x^2+y^2+2xy\sin\alpha$ is :

दिया है कि प्रतिलोम त्रिकोणमति फलन केवल मुख्य मान मानता है। माना [–1,1] में दो वास्तविक संख्याऐं x तथा y इस प्रकार है कि $\cos^{-1}x-\sin^{-1}y=\alpha$, $\frac{-\pi}{2}\le\alpha\le\pi$ तो $x^2+y^2+2xy\sin\alpha$ का न्यूनतम मान है :

(A) –1 (B) $\frac{1}{2}$ (C) $\frac{-1}{2}$ (D) 0

Ans. D

Sol. $\cos^{-1}x-\left(\frac{\pi}{2}-\cos^{-1}y\right)=\alpha$

$\cos^{-1}x+\cos^{-1}y=\frac{\pi}{2}+\alpha$

$\alpha\in\left[-\frac{\pi}{2},\pi\right],\frac{\pi}{2}+\alpha\in\left[0,\frac{3\pi}{2}\right]$

$\cos^{-1}\left(xy-\sqrt{1-x^2}\sqrt{1-y^2}\right)=\frac{\pi}{2}+\alpha$

$xy - \sqrt{1-x^2}\sqrt{1-y^2} = -\sin\alpha$

$(xy + \sin\alpha)^2 = (1 - x^2)(1 - y^2)$

$x^2y^2 + 2xy \sin a + \sin^2 a = 1 - x^2 - y^2 + x^2y^2$

$x^2 + y^2 + 2xy \sin\alpha = 1 - \sin^2\alpha$

$x^2 + y^2 + 2xy \sin\alpha = \cos^2\alpha$

Minimum value of $\cos^2\alpha = 0$

At $\alpha = \frac{\pi}{2}$

78. Let $x = \frac{m}{n}$ (m, n are co-prime natural numbers) be a solution of the equation $\cos\left(2\sin^{-1}x\right) = \frac{1}{9}$ and let α, β ($\alpha > \beta$) be the roots of the equation $mx^2 - nx - m + n = 0$. Then the point (α,β) lies on the line :

माना समीकरण $\cos\left(2\sin^{-1}x\right) = \frac{1}{9}$ का एक हल $x = \frac{m}{n}$ (m, n असहभाज्य घनपूर्णांक हैं) है और माना समीकरण

$mx^2 - nx - m + n = 0$ के मूल α, β ($\alpha > \beta$) हैं। तो बिन्दु (α,β) किस रेखा पर है :

(A) $5x + 8y = 9$ (B) $5x - 8y = -9$ (C) $3x + 2y = 2$ (D) $3x - 2y = -2$

Ans. A

Sol. $\sin^{-1}x = \theta \in I^{st}$ quadrant

$\cos 2\theta = \frac{1}{9}$

$\Rightarrow 1 - 2\sin^2\theta = \frac{1}{9}$

$\Rightarrow \sin^2\theta = \frac{4}{9}$

$\Rightarrow x^2 = \frac{4}{9} \Rightarrow x = \frac{2}{3} = \frac{m}{n}$

So $mx^2 - nx - m + n = 0$

$\Rightarrow 2x^2 - 3x + 1 = 0$

roots are $1, \frac{1}{2}$

$\alpha = 1, \beta = \frac{1}{2}$

So $(\alpha,\beta) = \left(1, \frac{1}{2}\right)$ lies or $5x + 8y = 9$

79. For $\alpha, \beta, \gamma \neq 0$, if $\sin^{-1}\alpha + \sin^{-1}\beta + \sin^{-1}\gamma = \pi$ and $(\alpha + \beta + \gamma)(\alpha - \gamma + \beta) = 3\alpha\beta$, then γ equals :

$\alpha, \beta, \gamma \neq 0$, के लिए, यदि $\sin^{-1}\alpha + \sin^{-1}\beta + \sin^{-1}\gamma = \pi$ है तथा $(\alpha + \beta + \gamma)(\alpha - \gamma + \beta) = 3\alpha\beta$ है, तो γ बराबर है :

(A) $\frac{1}{\sqrt{2}}$ (B) $\frac{\sqrt{3}}{2}$ (C) $\sqrt{3}$ (D) $\frac{\sqrt{3}-1}{2\sqrt{2}}$

Ans. B

Sol. Let $\sin^{-1}\alpha = A$, $\sin^{-1}\beta = B$ and $\sin^{-1}\gamma = C$

$\therefore A + B + C = \pi$

Also, $(\alpha + \beta + \gamma)(\alpha - \gamma + \beta) = 3\alpha\beta$

$\Rightarrow (\alpha+\beta)^2 - \gamma^2 = 3\alpha\beta$

$\Rightarrow \alpha^2 + \beta^2 - \gamma^2 = \alpha\beta$

$\Rightarrow \frac{\alpha^2+\beta^2-\gamma^2}{2\alpha\beta} = \frac{1}{2}$

$\Rightarrow \cos C = \frac{1}{2} \Rightarrow C = 60°$

$\therefore \sin C = \frac{\sqrt{3}}{2} = \gamma$

80. If $a = \sin^{-1}(\sin(5))$ and $b = \cos^{-1}(\cos(5))$, then $a^2 + b^2$ is equal to :

यदि $a = \sin^{-1}(\sin(5))$ तथा $b = \cos^{-1}(\cos(5))$, तो $a^2 + b^2$ बराबर है :

(A) $4\pi^2 + 25$ (B) $4\pi^2 - 20\pi + 50$ (C) $8\pi^2 - 40\pi + 50$ (D) 25

Ans. C

Sol. $a = \sin^{-1}(\sin 5) = 5 - 2\pi$

and $b = \cos^{-1}(\cos 5) = 2\pi - 5$

$\therefore a^2 + b^2 = (5-2\pi)^2 + (2\pi-5)^2$

$= 8\pi^2 - 40\pi + 50$

81. $\tan^{-1}\left(\frac{1+\sqrt{3}}{3+\sqrt{3}}\right) + \sec^{-1}\left(\sqrt{\frac{8+4\sqrt{3}}{6+3\sqrt{3}}}\right)$ is equal to :

$\tan^{-1}\left(\frac{1+\sqrt{3}}{3+\sqrt{3}}\right) + \sec^{-1}\left(\sqrt{\frac{8+4\sqrt{3}}{6+3\sqrt{3}}}\right)$ बराबर है :

(A) $\frac{\pi}{6}$ (B) $\frac{\pi}{2}$ (C) $\frac{\pi}{4}$ (D) $\frac{\pi}{3}$

Ans. D

Sol. $E = \tan^{-1}\left(\frac{1+\sqrt{3}}{\sqrt{3}(\sqrt{3}+1)}\right)+\sec^{-1}\left(\frac{16+8\sqrt{3}}{12+6\sqrt{3}}\right)^{1/2}$

$\because \frac{16+8\sqrt{3}}{12+6\sqrt{3}} = \frac{4(\sqrt{3}+1)^2}{(3+\sqrt{3})^2}$

$\therefore \left(\frac{16+8\sqrt{3}}{12+6\sqrt{3}}\right)^{1/2} = \frac{2(\sqrt{3}+1)}{3+\sqrt{3}} = \frac{2}{\sqrt{3}}$

$E = \tan^{-1}\left(\frac{1}{\sqrt{3}}\right)+\sec^{-1}\left(\frac{2}{\sqrt{3}}\right) = \tan^{-1}\left(\frac{1}{\sqrt{3}}\right)+\cos^{-1}\left(\frac{\sqrt{3}}{2}\right)$

$E = \tan^{-1}\left(\frac{1}{\sqrt{3}}\right)+\cos^{-1}\left(\frac{\sqrt{3}}{2}\right)$

$= \frac{\pi}{6}+\frac{\pi}{6} = \frac{\pi}{3}$

82. If $\sin^{-1}\frac{\alpha}{17}+\cos^{-1}\frac{4}{5}-\tan^{-1}\frac{77}{36}$, $0<\alpha<13$ then $\sin^{-1}(\sin\alpha)+\cos^{-1}(\cos\alpha)$ is equal to

यदि $\sin^{-1}\frac{\alpha}{17}+\cos^{-1}\frac{4}{5}-\tan^{-1}\frac{77}{36}$, $0<\alpha<13$ है, तो $\sin^{-1}(\sin\alpha)+\cos^{-1}(\cos\alpha)$ बराबर है :

(A) 16 (B) 0 (C) π (D) $16-5\pi$

Ans. C

Sol. $\cos^{-1}\frac{4}{5} = \tan^{-1}\frac{3}{4}$

$\therefore \sin^{-1}\frac{\alpha}{17} = \tan^{-1}\frac{77}{36}-\tan^{-1}\frac{3}{4} = \tan^{-1}\left(\frac{\frac{77}{36}-\frac{3}{4}}{1+\frac{77}{36}.\frac{3}{4}}\right)$

$\sin^{-1}\frac{\alpha}{17} = \tan^{-1}\frac{8}{15} = \sin^{-1}\frac{8}{17}$

$\Rightarrow \frac{\alpha}{17} = \frac{8}{17} \Rightarrow \alpha = 8$

$\therefore \sin^{-1}(\sin 8)+\cos^{-1}(\cos 8)$

$= 3\pi - 8 + 8 - 2\pi$

$= \pi$

83. Let $a_1 = 1, a_2, a_3, a_4, \ldots\ldots$ be consecutive natural numbers.

Then $\tan^{-1}\left(\frac{1}{1+a_1a_2}\right)+\tan^{-1}\left(\frac{1}{1+a_2a_3}\right)+\ldots\ldots+\tan^{-1}\left(\frac{1}{1+a_{2021}a_{2022}}\right)$ is equal to

माना $a_1 = 1, a_2, a_3, a_4, \ldots\ldots$ क्रमागत धन पूर्णांक है। तब

$\tan^{-1}\left(\frac{1}{1+a_1a_2}\right)+\tan^{-1}\left(\frac{1}{1+a_2a_3}\right)+\ldots\ldots+\tan^{-1}\left(\frac{1}{1+a_{2021}a_{2022}}\right)$ बराबर है :

(A) $\tan^{-1}(2022)-\frac{\pi}{4}$ (B) $\frac{\pi}{4}-\cot^{-1}(2022)$ (C) $\frac{\pi}{4}-\tan^{-1}(2022)$ (D) $\cot^{-1}(2022)-\frac{\pi}{4}$

Ans. A,B

Sol. $a_2 - a_1 = a_3 - a_2 = \ldots\ldots = a_{2022} - a_{2021} = 1.$

$\therefore \tan^{-1}\left(\frac{a_2-a_1}{1+a_1a_2}\right)+\tan^{-1}\left(\frac{a_3-a_2}{1+a_2a_3}\right)+\ldots\ldots+\tan^{-1}\left(\frac{a_{2022}-a_{2021}}{1+a_{2021}a_{2022}}\right)$

$\left[\left(\tan^{-1}a_2\right)-\tan^{-1}a_1\right]+\left[\tan^{-1}a_3-\tan^{-1}a_2\right]+\ldots\ldots+\left[\tan^{-1}a_{2022}-\tan^{-1}a_{2021}\right]$

$= \tan^{-1} a_{2022} - \tan^{-1} a_1$

$= \tan^{-1}(2022) - \tan^{-1} 1 = \tan^{-1} 2022 - \frac{\pi}{4}$ **(Option 3)**

$=\left(\frac{\pi}{2}-\cot^{-1}(2022)\right)-\frac{\pi}{4}$

$=\frac{\pi}{4}-\cot^{-1}(2022)$ **(Option 1)**

84. Let $(a,b)\subset(0,2\pi)$ be the largest interval for which $\sin^{-1}(\sin\theta)-\cos^{-1}(\sin\theta)>0$, $\theta\in(0,2\pi)$, holds. If $\alpha x^2+\beta x+\sin^{-1}\left(x^2-6x+10\right)+\cos^{-1}\left(x^2-6x+10\right)=0$ and $\alpha-\beta=b-a$, then α is equal to :

माना $(a,b)\subset(0,2\pi)$ सबसे बड़ा अंतराल है, जिसके लिए $\sin^{-1}(\sin\theta)-\cos^{-1}(\sin\theta)>0$, $\theta\in(0,2\pi)$ है। यदि $\alpha x^2+\beta x+\sin^{-1}\left(x^2-6x+10\right)+\cos^{-1}\left(x^2-6x+10\right)=0$ तथा $\alpha-\beta=b-a$ है, तो α बराबर है :

(A) $\frac{\pi}{48}$ (B) $\frac{\pi}{16}$ (C) $\frac{\pi}{8}$ (D) $\frac{\pi}{12}$

Ans. D

Sol. $\sin^{-1}\sin\theta-\left(\frac{\pi}{2}-\sin^{-1}\sin\theta\right)>0$

$\Rightarrow \sin^{-1}\sin\theta > \frac{\pi}{4}$

$\Rightarrow \sin\theta > \frac{1}{\sqrt{2}}$

So, $\theta \in \left(\frac{\pi}{4}, \frac{3\pi}{4}\right)$

$\theta \in \left(\frac{\pi}{4}, \frac{3\pi}{4}\right) = (a, b)$

$\Rightarrow b - a = \frac{\pi}{2} = \alpha - \beta$

$\Rightarrow \beta = \alpha - \frac{\pi}{2}$

$\Rightarrow \alpha x^2 + \beta x + \sin^{-1}\left[(x-3)^2 + 1\right] + \cos^{-1}\left[(x-3)^2 + 1\right] = 0$

$x = 3, 9\alpha + 3\beta + \frac{\pi}{2} + 0 = 0$

$\Rightarrow 9\alpha + 3\left(\alpha - \frac{\pi}{2}\right) + \frac{\pi}{2} = 0$

$\Rightarrow 12\alpha - \pi = 0$

$\alpha = \frac{\pi}{12}$

85. Let S be the set of all solutions of the equation $\cos^{-1}(2x) - 2\cos^{-1}\left(\sqrt{1-x^2}\right) = \pi$, $x \in \left[-\frac{1}{2}, \frac{1}{2}\right]$. Then $\sum_{x\in S} 2\sin^{-1}\left(x^2 - 1\right)$ is equal to :

माना समीकरण $\cos^{-1}(2x) - 2\cos^{-1}\left(\sqrt{1-x^2}\right) = \pi$, $x \in \left[-\frac{1}{2}, \frac{1}{2}\right]$ के सभी हलों का समुच्चय S है। तब $\sum_{x\in S} 2\sin^{-1}\left(x^2 - 1\right)$ का मान है :

(A) 0 (B) $\pi - \sin^{-1}\left(\frac{\sqrt{3}}{4}\right)$ (C) $\frac{-2\pi}{3}$ (D) $\pi - 2\sin^{-1}\left(\frac{\sqrt{3}}{4}\right)$

Ans. C

Sol. $\cos^{-1}(2x) = \pi + 2\cos^{-1}\left(\sqrt{1-x^2}\right)$

LHS = [0, π]

For equation to be meaningful

$\cos^{-1} 2x = \pi$ and $\cos^{-1}\left(\sqrt{1-x^2}\right) = 0$

$x = \frac{-1}{2}$ and $x = 0$

which is not possible

$\therefore\ x \in \phi$

Now $\Sigma(x) = 0$

$\therefore$ Sum over empty set is always 0

86. If $y = \cos\left(\frac{\pi}{3} + \cos^{-1}\frac{x}{2}\right)$, then $(x-y)^2 + 3y^2$ is equal to _____.

यदि $y = \cos\left(\frac{\pi}{3} + \cos^{-1}\frac{x}{2}\right)$ है, तो $(x-y)^2 + 3y^2$ बराबर है _____.

Ans. 3

Sol. $y = \cos\left(\frac{\pi}{3} + \cos^{-1}\frac{x}{2}\right)$

$= \cos\left(\frac{\pi}{3}\right)\cos\left(\cos^{-1}\left(\frac{x}{2}\right)\right) - \sin\left(\frac{\pi}{3}\right)\sin\left(\cos^{-1}\left(\frac{x}{2}\right)\right)$

$= \frac{1}{2}\cdot\frac{x}{2} - \frac{\sqrt{3}}{2}\cdot\sqrt{1-\frac{x^2}{4}}$

$\Rightarrow 4y = x - \sqrt{3}\sqrt{4-x^2}$

$\Rightarrow (4y-x)^2 = 3(4-x^2)$

$\Rightarrow 16y^2 + x^2 - 8xy = 12 - 3x^2$

$x^2 + 4y^2 - 2xy = 3$

$(x-y)^2 + 3y^2 = 3$

87. Let $S = \{x : \cos^{-1} x = \pi + \sin^{-1} x + \sin^{-1}(2x+1)\}$. Then $\sum_{x\in S}(2x-1)^2$ is equal to _____.

माना कि $S = \{x : \cos^{-1} x = \pi + \sin^{-1} x + \sin^{-1}(2x+1)\}$ है, तब $\sum_{x\in S}(2x-1)^2$ बराबर है।_____.

Ans. 5

Sol. $\cos^{-1}x = \pi + \sin^{-1}x + \sin^{-1}(2x+1)$

$-\frac{\pi}{2} = 2\sin^{-1} x + \sin^{-1}(2x+1)$

$$-\frac{\pi}{2}-\sin^{-1}x=\sin^{-1}x+\sin^{-1}(2x+1)$$

$$-\sin\left(\frac{\pi}{2}+\sin^{-1}x\right)=\sin\left(\sin^{-1}x+\sin^{-1}(2x+1)\right)$$

$$-\cos(\sin^{-1}x)=x\cos(\sin^{-1}(2x+1))+\cos(\sin^{-1}x)(2x+1)$$

$$-\sqrt{1-x^2}=x\sqrt{1-(2x+1)^2}+(2x+1)\sqrt{1-x^2}$$

$$x^3-2x-1=0$$

$$(x+1)(x^2-x-1)=0$$

$x=-1$ (false) or $x^2-x-1=0$

$$x=\frac{1\pm\sqrt{1+4}}{2}$$

$$x=\frac{1\pm\sqrt{5}}{2}$$

$$2x-1=\pm\sqrt{5}$$

$$2x-1=\sqrt{5}$$

$$\sum_{x\in S}(2x-1)^2=5$$

88. Let the inverse trigonometric functions take principal values. The number of real solutions of the equation $2\sin^{-1}x+3\cos^{-1}x=\frac{2\pi}{5}$ is _____.

माना प्रतिलोम त्रिकोणमितीय फलन मुख्य मान लेते हैं। तो समीकरण $2\sin^{-1}x+3\cos^{-1}x=\frac{2\pi}{5}$ के वास्तविक हलों की संख्या है _____.

Ans. 0

Sol. $2\sin^{-1}x+2\cos^{-1}x+\cos^{-1}x=\frac{2\pi}{5}$

$$2\left(\frac{\pi}{2}\right)+\cos^{-1}x=\frac{2\pi}{5}$$

$$\cos^{-1}x=\frac{2\pi}{5}-\pi$$

$$\cos^{-1}x=\frac{-3\pi}{5}$$

But $\cos^{-1}x\in[0,\pi]$

$\therefore$ No real value of x

No solution

89. If $S=\left\{x\in\mathbb{R}:\sin^{-1}\left(\frac{x+1}{\sqrt{x^2+2x+2}}\right)-\sin^{-1}\left(\frac{x}{\sqrt{x^2+1}}\right)=\frac{\pi}{4}\right\}$, then

$\sum_{x\in s}\left(\sin\left(\left(x^2+x+5\right)\frac{\pi}{2}\right)-\cos\left(\left(x^2+x+5\right)\pi\right)\right)$ is equal to _____.

यदि $S=\left\{x\in\mathbb{R}:\sin^{-1}\left(\frac{x+1}{\sqrt{x^2+2x+2}}\right)-\sin^{-1}\left(\frac{x}{\sqrt{x^2+1}}\right)=\frac{\pi}{4}\right\}$, है, तो

$\sum_{x\in s}\left(\sin\left(\left(x^2+x+5\right)\frac{\pi}{2}\right)-\cos\left(\left(x^2+x+5\right)\pi\right)\right)$ बराबर है _____.

Ans. 4

Sol. $\sin^{-1}\left(\frac{x+1}{\sqrt{x^2+2x+2}}\right)-\sin^{-1}\left(\frac{x}{\sqrt{x^2+1}}\right)=\frac{\pi}{4}$

$\tan^{-1}(x+1)-\tan^{-1}x=\frac{\pi}{4}\Rightarrow\tan^{-1}\left(\frac{(x+1)-x}{1+(x+1)x}\right)=\frac{\pi}{4}$

$\tan^{-1}\left(\frac{1}{1+x^2+x}\right)=\frac{\pi}{4}\Rightarrow\frac{1}{1+x^2+x}=1$

$x^2+x+1=1\Rightarrow x^2+x=0$

$\Rightarrow x=0,-1\Rightarrow S=\{0,-1\}$

Now, $\sum_{x\in S}\left(\sin\left(\left(x^2+x+5\right)\right)\frac{\pi}{2}\right)-\cos\left(\left(x^2+x+5\right)\pi\right)$

$=\left(\sin\frac{5\pi}{2}-\cos5\pi\right)+\left(\sin\frac{5\pi}{2}-\cos5\pi\right)=(1+1)+(1+1)=4$